



HELLO

*PIZZA SALES EXPLORATORY DATA
ANALYSIS USING MY SQL*



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RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED.

SELECT

COUNT(order_id) **AS** Total_Orders

FROM

orders;

THE BUSINESS PROCESSED 21,350 CUSTOMER ORDERS, INDICATING STRONG SALES ACTIVITY AND CUSTOMER DEMAND THROUGHOUT THE ANALYZED PERIOD.



	Total_Orders
▶	21350

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CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES.



SELECT

ROUND(SUM(order_details.quantity * pizzas.price),2) AS Total_revenue

FROM

order_details

JOIN

pizzas ON pizzas.pizza_id = order_details.pizza_id;

THE PIZZA BUSINESS GENERATED APPROXIMATELY
8,17,860.05/- IN TOTAL REVENUE, SHOWING STRONG
OVERALL BUSINESS PERFORMANCE AND CUSTOMER
PURCHASING ACTIVITY.

	Total_revenue
▶	817860.05



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IDENTIFY THE HIGHEST-PRICED PIZZA.



```
SELECT
    pt.`name`, pz.price
FROM
    pizza_types pt
    JOIN
    pizzas pz ON pt.pizza_type_id = pz.pizza_type_id
ORDER BY pz.price DESC
LIMIT 1;
```

	name	price
▶	The Greek Pizza	35.95

THE GREEK PIZZA IS THE
HIGHEST-PRICED ITEM ON THE
MENU, SUGGESTING IT MAY BE
POSITIONED AS A PREMIUM
PRODUCT TARGETING HIGH-VALUE
CUSTOMERS.



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○ IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED.



```
SELECT
    pz.size, COUNT(od.Order_details_id) AS Order_count
FROM
    pizzas pz
    JOIN
    order_details od ON pz.pizza_id = od.pizza_id
GROUP BY pz.size
ORDER BY Order_count DESC;
```

*LARGE-SIZED PIZZAS DOMINATE CUSTOMER
ORDERS, INDICATING CUSTOMERS PREFER
SHARING PORTIONS OR PERCEIVE LARGER PIZZAS
AS OFFERING BETTER VALUE FOR MONEY.*



	size	Order_count
▶	L	18526
	M	15385
	S	14137
	XL	544
	XXL	28

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LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.



```
SELECT
    pt.`name` AS Pizza_name, SUM(od.Quantity) QTY
FROM
    pizza_types pt
    JOIN
    pizzas pz ON pt.pizza_type_id = pz.pizza_type_id
    JOIN
    order_details od ON od.pizza_id = pz.pizza_id
GROUP BY pt.`name`
ORDER BY QTY DESC
LIMIT 5;
```

*CLASSIC AND CHICKEN-BASED
PIZZAS ARE AMONG THE MOST
FREQUENTLY ORDERED ITEMS,
SHOWING STRONG CUSTOMER
PREFERENCE TOWARD FAMILIAR
AND PROTEIN-RICH FLAVORS.*

	Pizza_name	QTY
▶	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371



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JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED.

```
SELECT
    pt.category, SUM(od.Quantity) AS QTY
FROM
    pizza_types pt
    JOIN
    pizzas pz ON pt.pizza_type_id = pz.pizza_type_id
    JOIN
    order_details od ON od.pizza_id = pz.pizza_id
GROUP BY pt.category
ORDER BY QTY DESC;
```

CLASSIC CATEGORY GENERATED THE HIGHEST ORDER QUANTITY, MAKING IT THE MOST POPULAR PIZZA CATEGORY AMONG CUSTOMERS.

CHICKEN AND SUPREME CATEGORIES ALSO CONTRIBUTED SIGNIFICANTLY, INDICATING DIVERSIFIED CUSTOMER TASTE PREFERENCES.

	category	QTY
▶	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050



NEXT >



DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.



```
SELECT
    HOUR(order_time) AS `Hour`, COUNT(order_id) AS Order_count
FROM
    orders
GROUP BY `Hour`
ORDER BY Order_count DESC;
```

*CUSTOMER ORDERS PEAK DURING LUNCH AND EVENING HOURS,
INDICATING HIGH DEMAND DURING MEAL TIMES AND SUGGESTING
OPTIMAL STAFFING SHOULD BE FOCUSED AROUND THESE
PERIODS.*

	Hour	Order_count
▶	12	2520
	13	2455
	18	2399
	17	2336
	19	2009
	16	1920
	20	1642
	14	1472
	15	1468
	11	1231
	21	1198
	22	663
	23	28
	10	8
	9	1



NEXT >



JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS.



```
SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category;
```

	category	COUNT(name)
▸	Chicken	6
	Classic	8
	Supreme	9
	Veggie	9



THE MENU MAINTAINS BALANCED CATEGORY DISTRIBUTION, OFFERING CUSTOMERS A WIDE VARIETY OF PIZZA CHOICES ACROSS DIFFERENT PREFERENCES.

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GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY.

ON AVERAGE, APPROXIMATELY
138 PIZZAS ARE ORDERED
DAILY, DEMONSTRATING STABLE
DAY-TO-DAY CUSTOMER
DEMAND.

```
• SELECT
    ROUND(AVG(Qty),0) Avg_pizzas_ordered_perday
FROM
    (SELECT
        ods.order_date, SUM(od.quantity) AS Qty
    FROM
        orders ods
    JOIN order_details od ON od.order_id = ods.Order_id
    GROUP BY ods.order_date) AS Order_qty;
```

	Avg_pizzas_ordered_perday
▶	138



NEXT >



DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.



```
SELECT
    pt.`name`, SUM(od.quantity * pz.price) AS Revenue
FROM
    pizza_types pt
    JOIN
    pizzas pz ON pz.pizza_type_id = pt.pizza_type_id
    JOIN
    order_details od ON pz.pizza_id = od.pizza_id
GROUP BY pt.`name`
ORDER BY Revenue DESC
LIMIT 3;
```

CHICKEN-BASED PIZZAS GENERATE THE HIGHEST REVENUE, SUGGESTING STRONG CUSTOMER WILLINGNESS TO SPEND MORE ON PREMIUM CHICKEN VARIETIES.

THESE PIZZAS COULD BE PRIORITIZED IN PROMOTIONAL CAMPAIGNS AND COMBO OFFERS TO MAXIMIZE REVENUE GROWTH.



	name	Revenue
▶	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5

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CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.

```
• SELECT
  pt.category,
  ROUND(SUM(od.quantity * pz.price) / (SELECT
    ROUND(SUM(order_details.quantity * pizzas.price),
      2) AS Total_revenue
    FROM
      order_details
      JOIN
        pizzas ON pizzas.pizza_id = order_details.pizza_id) *100,2) AS `Revenue%`
FROM
  pizza_types pt
  JOIN
    pizzas pz ON pz.pizza_type_id = pt.pizza_type_id
  JOIN
    order_details od ON pz.pizza_id = od.pizza_id
GROUP BY pt.category
ORDER BY `Revenue%` DESC;
```

REVENUE CONTRIBUTION IS RELATIVELY BALANCED ACROSS CATEGORIES, REDUCING OVERDEPENDENCE ON A SINGLE PRODUCT CATEGORY.

	category	Revenue%
▶	Classic	26.91
	Supreme	25.46
	Chicken	23.96
	Veggie	23.68

THE CLASSIC CATEGORY CONTRIBUTES THE HIGHEST SHARE OF REVENUE, REINFORCING ITS IMPORTANCE TO THE BUSINESS.



NEXT >



ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.



```
• SELECT order_date, ROUND(SUM(Revenue)
  OVER(ORDER BY order_date),2)
  AS Cum_Revenue
FROM
  (SELECT odr.order_date,
    SUM(od.quantity * pz.price) AS Revenue
  FROM ORDERS odr
    JOIN order_details od
  ON od.Order_id = odr.order_id
    JOIN pizzas pz
  ON pz.pizza_id = od.pizza_id
  GROUP BY odr.order_date) AS Sales ;
```

order_date	Cum_Revenue
2015-01-01	2713.85
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05

CUMULATIVE REVENUE SHOWS CONSISTENT GROWTH OVER TIME, INDICATING STEADY CUSTOMER PURCHASING BEHAVIOR AND SUSTAINED BUSINESS PERFORMANCE.



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DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY.

```
• SELECT category, name, Revenue, RN
FROM
  (SELECT category, name, Revenue,
   RANK() OVER(PARTITION BY category
    ORDER BY Revenue DESC) AS RN
  FROM
    (SELECT pzt.category, pzt.name,
     ROUND(SUM((od.quantity) * pz.price),2)
     AS Revenue
    FROM pizzas pz
    JOIN order_details od
      ON pz.pizza_id = od.pizza_id
    JOIN pizza_types pzt
      ON pzt.pizza_type_id = pz.pizza_type_id
    GROUP BY pzt.category, pzt.name) AS A
  ) AS B
WHERE RN <= 3 ;
```

EACH PIZZA CATEGORY HAS CLEAR TOP-PERFORMING PRODUCTS, WHICH CAN HELP THE BUSINESS IDENTIFY HIGH-VALUE MENU ITEMS FOR MARKETING AND INVENTORY PRIORITIZATION.

	category	name	Revenue	RN
▶	Chicken	The Thai Chicken Pizza	43434.25	1
	Chicken	The Barbecue Chicken Pizza	42768	2
	Chicken	The California Chicken Pizza	41409.5	3
	Classic	The Classic Deluxe Pizza	38180.5	1
	Classic	The Hawaiian Pizza	32273.25	2
	Classic	The Pepperoni Pizza	30161.75	3
	Supreme	The Spicy Italian Pizza	34831.25	1
	Supreme	The Italian Supreme Pizza	33476.75	2
	Supreme	The Sicilian Pizza	30940.5	3
	Veggie	The Four Cheese Pizza	32265.7	1
	Veggie	The Mexicana Pizza	26780.75	2
	Veggie	The Five Cheese Pizza	26066.5	3

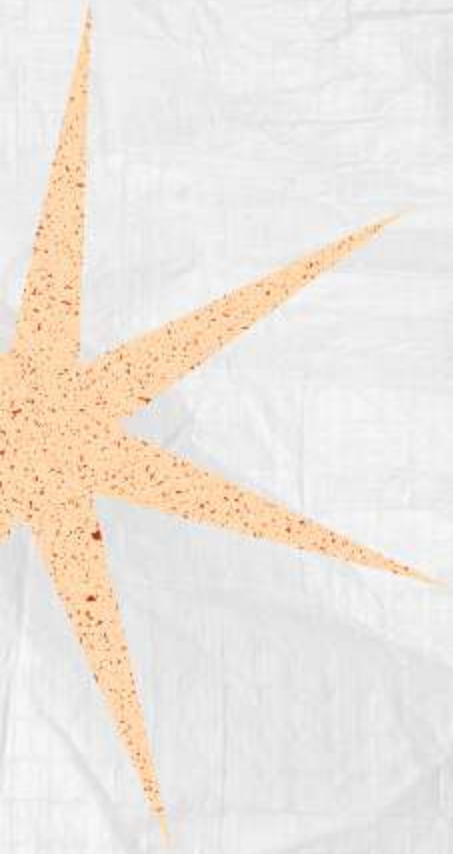
THAI CHICKEN PIZZA EMERGED AS ONE OF THE STRONGEST PERFORMERS OVERALL, PARTICULARLY WITHIN THE CHICKEN CATEGORY.



NEXT >



THANK YOU!



BY PUNEETH



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